FIITJEE

ALL INDIA TEST SERIES

PART TEST – II

JEE (Advanced)-2018-19

PAPER –1

TEST DATE: 18-11-2018

Time Allotted: 3 Hours **Maximum Marks: 183**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-C.**

  Section-A (01– 07, 19 – 25, 37 - 43) *contains 21 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 mark*** *for wrong answer*
  Partial Marks **+1** for each correct option provided no incorrect options is selected.

  Section-A (08 – 13, 26 – 31, 44 - 49) *contains 18 questions. Each of 2 Tables with 3 Columns and 4 Rows has three questions. Column 1 will be with 4 rows designated (I), (II), (III) and (IV). Column 2 will be with 4 rows designated (i), (ii), (iii) and (iv). Column 3 will be with 4 rows designated (P), (Q), (R) and (S).*
  Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-C (14 – 18, 32 – 36, 50 - 54) *contains 15 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Physics

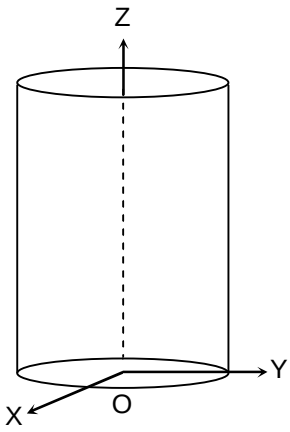
PART – I

SECTION – A
(More Than One Correct Type)

This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

1. The electric field in the space is given by $\vec{E} = E_0\left(x\hat{i} + y\hat{j} + z\hat{k}\right)$.

Consider a right circular cylindrical surface whose radius is 'a' and height 'h', as shown in the figure.
Now choose the correct option(s).



(A) The electric flux through lower circular base is zero.

(B) The electric flux through upper circular top is $\pi a^2 h E_0$

(C) The electric flux through lateral surface is $2\pi a^2 h E_0$

(D) The total electric flux through cylindrical surface is $4\pi a^2 h E_0$

Ans. **A, B, C**

Sol. Let, S_1, S_2 and S_3 be the surface areas of the lower circular base, upper circular top and lateral surface of the cylinder respectively.

$$\phi_1 = \int_{S_1} \vec{E}.d\vec{S} = 0 \qquad \ldots(1)$$

$$\phi_2 = \int_{S_2} \vec{E}.d\vec{S} = \pi a^2 h E_0 \qquad \ldots(2)$$

$$\phi_3 = \int_{S_3} \vec{E}.d\vec{S} = E_0 \int_{S_3} \vec{r}.d\vec{S} \qquad \ldots(3)$$

$$\vec{r}.d\vec{S} = \left(x\hat{i} + y\hat{j} + z\hat{k}\right).\left(\frac{x\hat{i} + y\hat{j}}{a}\right)dS = \frac{x^2 + y^2}{a}dS$$

Putting this value in equation (3), we have

$$\phi_3 = aE_0 \int_{S_3} dS = aE_0 \times 2\pi ah = 2\pi a^2 h E_0$$

$$\phi = \text{Total flux} = \phi_1 + \phi_2 + \phi_3 = 3\pi a^2 h E_0$$

Second Method

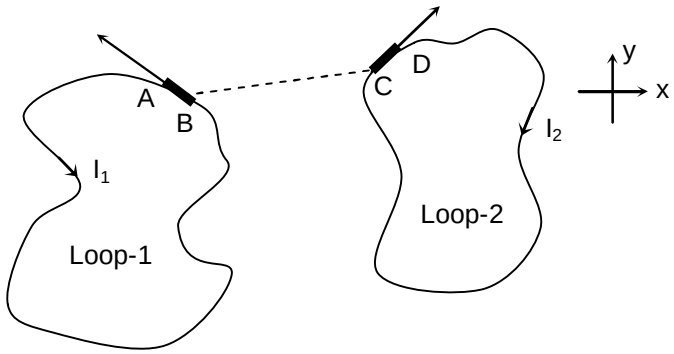
$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \Rightarrow 3E_0 = \frac{\rho}{\varepsilon_0} \Rightarrow \rho = 3\varepsilon_0 E_0$$

Q_{en} = Total charge enclosed within the cylinder = $\pi a^2 h\rho$

$= 3\pi a^2 h\varepsilon_0 E_0$

Total flux = $\dfrac{Q_{en}}{\varepsilon_0} = 3\pi a^2 hE_0$

2. Consider two current carrying loops in x-y plane, carrying current I_1 and I_2 respectively as shown in the figure.



Now consider two arbitrary elementary current elements AB & CD of lengths $d\ell_1$ and $d\ell_2$ on the loop-1 and loop-2 respectively. Let $d\vec{F}_{12}$ be the force which the current carrying element $I_2 d\vec{\ell}_2$ exerts on the current carrying element $I_1 d\vec{\ell}_1$ & $\vec{F}_{12}$ represent the net force on 1st loop by the 2nd loop, developed due to interaction between current carrying loops. Similarly $d\vec{F}_{21}$ be the force which the current carrying element $I_1 d\vec{\ell}_1$ exerts on the current carrying element $I_2 d\vec{\ell}_2$ & $\vec{F}_{21}$ represent the net force on 2nd loop by the 1st loop, developed due to interaction between current carrying loops. Choose the correct option(s)

(A) $d\vec{F}_{12} + d\vec{F}_{21} = \vec{0}$

(B) $d\vec{F}_{12} + d\vec{F}_{21} \neq \vec{0}$

(C) $\vec{F}_{12} + \vec{F}_{21} = \vec{0}$

(D) $\vec{F}_{12} + \vec{F}_{21} \neq \vec{0}$

Ans. B, C

Sol. $d\vec{B}_{12} =$ Magnetic field at any point on the elementary element AB due to current carrying element $I_2 d\vec{\ell}_2$.

$$d\vec{B}_{12} = \frac{\mu_0}{4\pi}\left(\frac{I_2 d\vec{\ell} \times \vec{r}_{12}}{r_{12}^3}\right)$$

$$d\vec{F}_{12} = I_1 d\vec{\ell}_1 \times d\vec{B}_{12} = \left(\frac{\mu_0}{4\pi}\right)\frac{I_1 d\vec{\ell}_1 \times \left(I_2 d\vec{\ell}_2 \times \vec{r}_{12}\right)}{r_{12}^3}$$

The force $d\vec{F}_{12}$ will be in the plane of loop and perpendicular to the current carrying element $I_1 d\vec{\ell}_i$ (say along $\hat{p}_{12}$).

Similarly $d\vec{F}_{21} = \dfrac{\mu_0}{4\pi}\dfrac{I_2 d\vec{\ell}_2 \times \left(I_1 d\vec{\ell}_1 \times \vec{r}_{21}\right)}{r_{21}^3}$ [Say along $\hat{p}_{21}$]

$\therefore$ $\hat{\rho}_{12}$ is not parallel to $\hat{\rho}_{21}$, in general

So, $d\vec{F}_{12} + d\vec{F}_{21} \neq \vec{0}$

But $\vec{F}_{12} = \dfrac{\mu_0 I_1 I_2}{4\pi} \int_{L_1} \int_{L_2} \dfrac{d\vec{\ell}_1 \times \left(d\vec{\ell}_2 \times \vec{r}_{12}\right)}{r_{12}^3}$

$\vec{F}_{21} = \dfrac{\mu_0 I_1 I_2}{4\pi} \int_{L_2} \int_{L1} \dfrac{d\vec{\ell}_2 \times \left(d\vec{\ell}_1 \times \vec{r}_{21}\right)}{r_{21}^3}$.

Hence, $\vec{F}_{12} + \vec{F}_{21} = \vec{0}$

3. Consider an isolated sphere of radius 'b', made of a material with permittivity ε_0 and electrical conductivity γ, whose volume charge density varies with time t and is given as

$$\rho = \begin{cases} \rho_0 e^{-t/\tau} & \text{if} \quad 0 \le r < a \\ 0 & \text{if} \quad a < r < \infty \end{cases}$$

Where $\tau = \dfrac{\varepsilon_0}{\gamma}$ and 0 < a < b. The spherical layer between radii 'a' and 'b' is neutral. Then choose the correct option(s).

(A) The surface charge density at outer surface of sphere is $\dfrac{a^3\rho_0}{8b^2}$ at $t = \tau \ell n 2$.

(B) The total current in the region 0 < r < a increases with the radius.

(C) The electric field in the region b < r < ∞ is independent of time.

(D) The total current in the region a < r < b is independent of radius.

Ans. B, C, D

Sol. $$Q(t) = \begin{cases} \dfrac{4\pi}{3} a^3 \rho_0 e^{-t/\tau} = Q_0 e^{-t/\tau} & \text{if} \quad 0 \le r < a \\ 0 & \text{if} \quad a < r < b \end{cases}$$

Here $Q_0 = \dfrac{4\pi}{3} a^3 \rho_0$

According Gauss' Law, we can write

$$E = \begin{cases} \dfrac{Q_0 r\, e^{-t/\tau}}{4\pi\varepsilon_0 a^3} & \text{if} \quad 0 \le r < a \\ \dfrac{Q_0\, e^{-t/\tau}}{4\pi\varepsilon_0 r^2} & \text{if} \quad a < r < b \\ \dfrac{Q_0}{4\pi\varepsilon_0 r^2} & \text{if} \quad b < r < \infty \end{cases}$$

Using conservation of charge we can write surface charge density at the outer surface of sphere as,

$$\sigma = \frac{Q_0 - Q(t)}{4\pi b^2} = \frac{Q_0}{4\pi b^2}\left(1 - e^{-t/\tau}\right)$$

If $t = \tau \ell n 2 \Rightarrow 1 - e^{-\ell n 2} = \dfrac{1}{2}$,

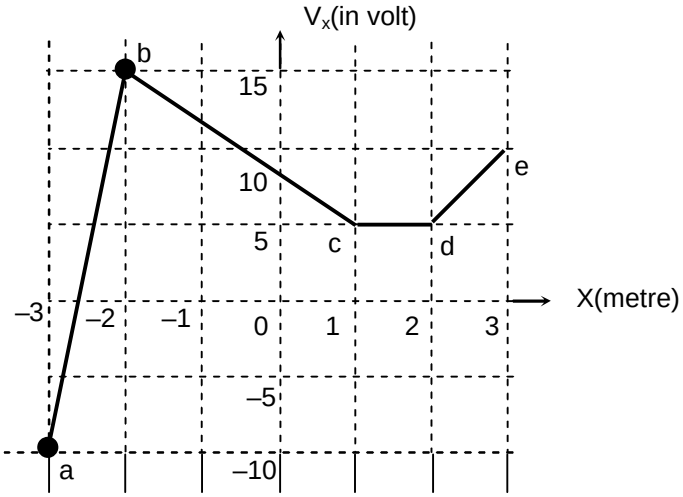
So, $\sigma = \frac{4}{3}\pi a^3 \rho_0 \times \frac{1}{4\pi b^2} \times \frac{1}{2} = \frac{a^3\rho_0}{6b^2}$

J = current density = γE

I = current = $J\pi r^2$

Hence in the region $0 \le r < a$, the volume is source of conduction current while the region a < r < b has no source of conduction current.

4. The variation of potential along x-axis is shown in the figure. The potential does not vary along the y-axis or z-axis. Ignore the behaviour at the end points of interval. Now choose the correct option(s) for given interval –3m < x < 3m.



(A) The greatest absolute value of E_x is 25 V/m.

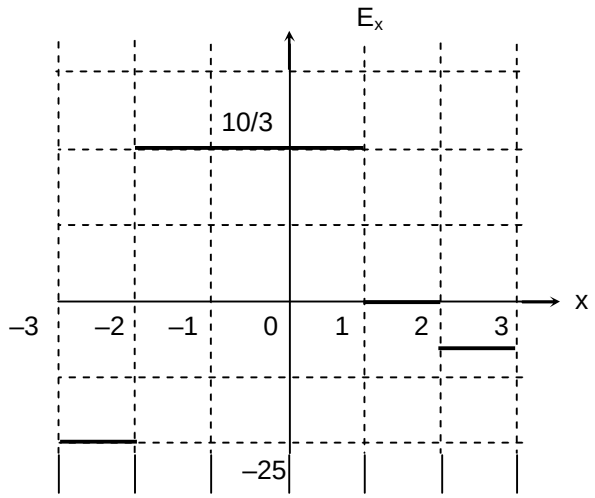
(B) An electron is released at point (– 1m, 0), it will move along positive x-axis.

(C) A proton is released at point (1.5 m, 0), it will move along positive x-axis.

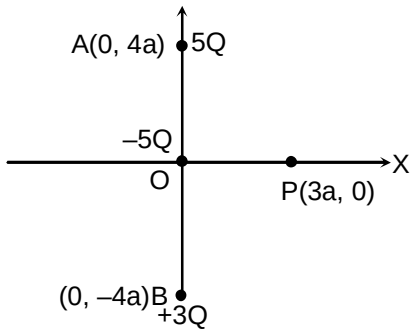
(D) An electron is released at point (– 2.5 m, 0), it will move along positive x-axis.

Ans. A, D

Sol.



5. Three point charges are kept at points A(0, 4a), O(0, 0) and B(0, –4a) as shown in the figure.



(A) Work was done on the external agent who assembled these charges in moving them from infinity to the point.

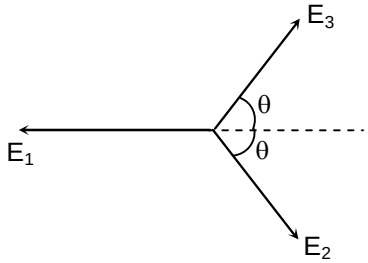
(B) The electric field at point P is $\vec{E}_p = E_X \hat{i} + E_y \hat{j}$, then E_x is negative and E_y is positive.

(C) The electric field at point P is $\vec{E}_p = E_x \hat{i} + E_y \hat{j}$, then both E_x and E_y are negative.

(D) A positive charge $\frac{975Q}{8}$ has to be put at point P to make the potential energy of system of all four point charges to be zero.

Ans. A, C

Sol. Potential energy of the system is negative so option A is correct



$$\theta = 53^\circ, E_1 = \frac{5KQ}{9}, E_2 = \frac{KQ}{5}, E_3 = \frac{3KQ}{25}$$

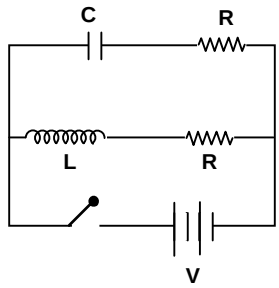
$$E_x = -\frac{5KQ}{9} + \frac{KQ}{5}\times\frac{3}{5} + \frac{3KQ}{25}\times\frac{3}{5} = \frac{-5KQ}{9} + \frac{24KQ}{125} < 0$$

$$E_y = \frac{-K\theta}{5}\times\frac{4}{5} + \frac{3K\theta}{25}\times\frac{4}{5} = \frac{-8K\theta}{125} < 0$$

$$\frac{-25Q^2}{4} + \frac{15Q^2}{8} + \frac{5xQ}{5} + \frac{-5Qx}{3} + \frac{-15Q^2}{4} + \frac{3Qx}{5} = 0$$

$$\frac{-65Q^2}{8} = \frac{8Qx}{5} - \frac{5Qx}{3} = 0 \Rightarrow \frac{(24-25)x}{15} = \frac{65Q}{8} \Rightarrow x = \frac{-65\times 15Q}{8}$$

6. In the circuit shown in figure, the switch S is closed at the moment t = 0. At the moment t = RCℓn2, the currents through both resistors are found to be 'I'. Then choose the correct option(s)



(A) $I = \frac{V}{R}$

(B) $I = \frac{V}{2R}$

(C) $R = \sqrt{\frac{L}{2C}}$

(D) $R = \sqrt{\frac{L}{C}}$

Ans. B, D

Sol. $I_1 = \frac{V}{R}\left(1 - e^{-\frac{Rt}{L}}\right)$, and $I_2 = \frac{V}{R}e^{-\frac{t}{RC}}$

According to question at $t = RC\ell n2$, we have

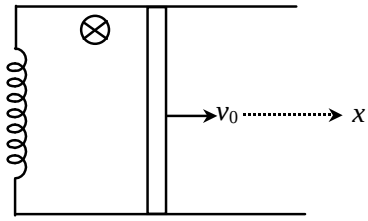
$I_1 = I_2 \Rightarrow \frac{V}{R}\left(1 - e^{-\frac{Rt}{L}}\right) = \frac{V}{R}e^{-\frac{t}{RC}} \Rightarrow 1 - e^{-\frac{Rt}{L}} = e^{-\frac{t}{RC}}$ .....(1)

Given $t = RC\ell n2 \Rightarrow e^{-\frac{t}{RC}} = e^{-\ell n2} = \frac{1}{2}$, Putting this value in equation (1), we have

$\Rightarrow R = \sqrt{\frac{L}{C}}$

$1 - e^{-\frac{Rt}{L}} = \frac{1}{2} \Rightarrow e^{-\frac{Rt}{L}} = \frac{1}{2} \Rightarrow \frac{Rt}{L} = \ell n2 \Rightarrow \frac{R}{L} \times RC\ell n2 = \ell n2$

7. A loop is formed by two fixed parallel conductors connected by a solenoid with inductance L = 2H and a conducting rod of mass m = 8 kg which can freely slide without friction, over the conductors. The conductors are located in a horizontal plane and in a uniform vertical magnetic field B = π Tesla directed into the plane of the paper. The distance between the conductors is ℓ = 2 m. At the moment t = 0, the rod is imparted an initial velocity v_0 = 2 m/s directed to the right. The resistance of loop is negligible. Choose the correct option(s)



(A) The rod will perform periodic motion, but not SHM

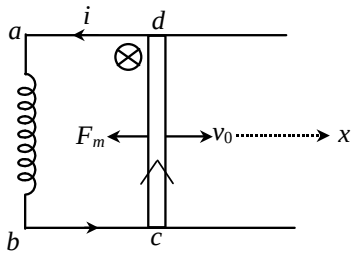
(B) The rod will perform SHM

(C) The time period of oscillation of rod is 4 s

(D) The amplitude of oscillation of rod is $\frac{4}{\pi}$ metre.

Ans. B, C, D

Sol. Suppose that at any instant of time, velocity of the rod is *v* towards right.



The current in the circuit is *i*. In the figure,

$$V_a - V_b = V_d - V_c$$

i.e. $Ldi = Bldx$

Integrating, we get $Li = Blx$

Magnetic force on the rod at this instant is

$$F_m = i\ell B = \frac{B^2\ell^2}{L}x \qquad \ldots (i)$$

Since, this force is in opposite direction of $\vec{v}$, so from Newton's second law we can write,

$$m\left(\frac{d^2x}{dt^2}\right) = -\frac{B^2l^2}{L}x$$

Comparing this with equation of SHM, i.e.

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 \Rightarrow x = A\sin\omega t \Rightarrow v = A\omega\cos\omega t \ldots (ii)$$

We have, $\omega = \frac{Bl}{\sqrt{mL}}$

So $T = \frac{2\pi}{\omega} = 2\pi\frac{\sqrt{mL}}{Bl} = 2\pi\frac{\sqrt{8\times 2}}{\pi\times 2} = 4\ s$

At t = 0, $v = v_0 \Rightarrow v_0 = A\omega \Rightarrow A = \frac{v_0}{\omega} = \frac{4}{\pi}m$

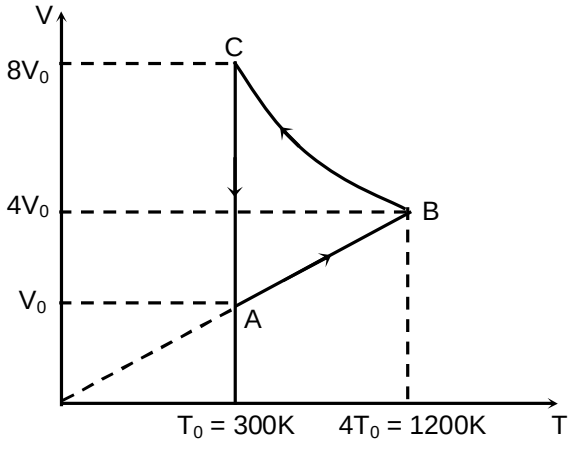
(Matching Type)
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer 8, 9 and 10 by appropriately matching the information given in the three columns of the following table.

Two moles of an ideal mono-atomic gas is taken through a cyclic process ABCA as shown in the V-T diagram. In the process BC, TV^2 = constant. Answer the questions by referring to the data in the table.

(Take $\ell n(2) = 0.7$)



Column –1 shows the processes. Column –2 shows the work done in that process and Column –3 shows the energy supplied in that process.

Column – 1		Column – 2		Column – 3	
(I)	$A \rightarrow B$	(i)	1800 R	(P)	–1800 R
(II)	$B \rightarrow C$	(ii)	–1260 R	(Q)	– 3060 R
(III)	$C \rightarrow A$	(iii)	900 R	(R)	4500 R
(IV)	$B \rightarrow C \rightarrow A \rightarrow B$	(iv)	1440 R	(S)	1440 R

8. Pick a combination in which work done is maximum

(A) (I) (i) (R)

(B) (II) (i) (P)

(C) (III) (iv) (S)

(D) (III) (ii) (R)

Ans. A

9. Pick a combination in which thermal energy supplied is maximum

(A) (II) (iii) (P)

(B) (IV) (iii) (S)

(C) (I) (i) (R)

(D) (I) (ii) (Q)

Ans. C

10. Pick a correct combination

(A) (I) (ii) (P)

(B) (I) (ii) (Q)

(C) (IV) (iv) (S)

(D) (II) (i) (Q)

Ans. C

Sol.8-10. The equation of the process BC is

$TV^2 = \text{constant}$

$PV^3 = \text{constant}$

Molar heat capacity of the gas in the process BC is

$\therefore \quad C = C_V + \dfrac{R}{1-x}$

$= \dfrac{3R}{2} - \dfrac{R}{2} = R$

The work done by the gas during the process BC is

$\therefore \quad \Delta W_{BC} = \dfrac{nR\Delta T}{1-x} = \dfrac{2R(-900)}{1-3} = 900R$

$\Delta Q_{AB} = nCp\Delta T = 2 \times \dfrac{5R}{2}(1200-300) = 4500R$

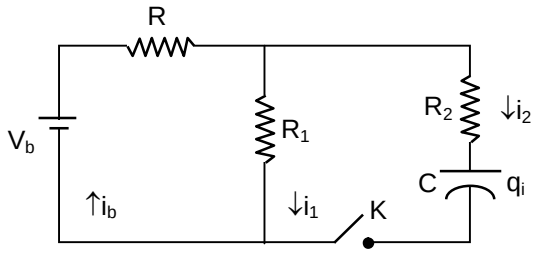
$\Delta Q_{BC} = nC\Delta T = 2 \times R(300-1200) = -1800R$

$\Delta Q_{CA} = \Delta W_{CA} = nRT_0 \ln\left(\dfrac{V_0}{8V_0}\right) = 2R300(-3\ell n2) = -1800R \times 0.7 = -1260R$

$\therefore \quad \Delta W_{cycle} = \Delta Q_{cycle} = \Delta Q_{AB} + \Delta Q_{BC} + \Delta Q_{CA} = 4500R - 1800R - 1260R = 1440\ R$

$\therefore$ Efficiency, $\eta = \dfrac{\Delta W_{cycle}}{\Delta Q_{supplied}} \times 100 = \dfrac{1440R}{4500R} \times 100 = 32\%$

Answer 11, 12 and 13 by appropriately matching the information given in the three columns of the following table.



q_i is the initial charge on the capacitor. The key K is closed at t = 0

Column – 1 Battery		Column – 2 Resistance		Column – 3 Capacitance & Initial charge	
(I)	V_b = 0 V	(i)	$R_1 = R_2 = 5\ \Omega$ $R = 10\ \Omega$	(P)	$C = 5\ \mu F$ $q_i = 0$
(II)	V_b = 30 V	(ii)	$R_1 = 5\ \Omega$ $R_2 = 10\ \Omega$ $R = 10\ \Omega$	(Q)	$C = 20\ \mu F$ $q_i = 0$
(III)	V_b = –30 V	(iii)	$R_1 = 10\ \Omega$ $R_2 = 5\ \Omega$ $R = 10\ \Omega$	(R)	$C = 5\ \mu F$ $q_i = 100\ \mu C$
(IV)	V_b = 60 V	(iv)	$R_1 = R_2 = R = 10\ \Omega$	(S)	$C = 20\ \mu F$ $q_i = 100\ \mu C$

11. In which of the following cases, will the current i_1 = 5/3 A, initially?

(A) (I) (i) (P)

(B) (II) (iv) (R)

(C) (II) (iv) (P)

(D) (I) (iii) (Q)

Ans. B

12. In which of the following cases will the net charge flowing through the capacitor be equal to 300 μC?

(A) (II) (i) (S)

(B) (I) (ii) (R)

(C) (IV) (i) (S)

(D) (III) (iii) (P)

Ans. C

13. In which case will the current through R decay exponentially to zero with a time constant of 0.2 ms?

(A) (IV) (ii) (Q)

(B) (I) (iii) (S)

(C) (II) (i) (R)

(D) (I) (iii) (Q)

Ans. B

Sol.11-13. $V_b - i_b R - i_1 R_1 = 0$

$-i_2 R_2 - q/C + i_1 R_1 = 0$

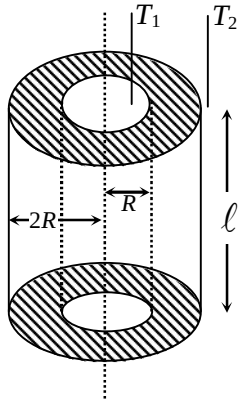
With $i_2 = \frac{dq}{dt}$

Substituting the given values, we get the result.

SECTION – C
(Single digit integer type)

This section contains **FIVE** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

14. Inner surface of a cylindrical shell of length ℓ and of material of thermal conductivity k is kept at constant temperature T_1 and outer surface of the cylinder is kept at constant temperature T_2 such that $(T_1 > T_2)$ as shown in figure. Heat flows from inner surface to outer surface radially outward. Inner and outer radii of the shell are R and 2R respectively. Due to lack of space this cylinder has to be replaced by a smaller cylinder of length $\frac{\ell}{2}$, inner and outer radii $\frac{R}{4}$ and R respectively and thermal conductivity of material nk. If rate of radial outward heat flow remains same for same temperatures of inner and outer surface i.e. T_1 and T_2, then find the value of n.

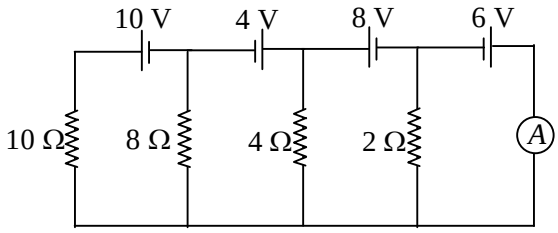


Ans. 4

Sol. Since rate of heat flow remains same in both the cases, so

$$\int_{R}^{2R}\frac{dx}{k2\pi x\ell}=\int_{\frac{R}{4}}^{R}\frac{dx}{nk(2\pi x)\frac{\ell}{2}} \Rightarrow nk = 4k \Rightarrow n = 4$$

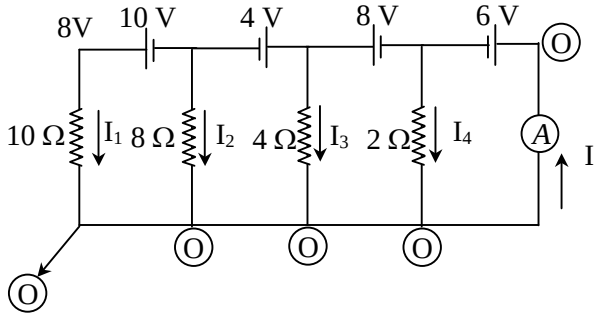
15. If the reading of the ideal ammeter connected in the given circuit is $n\left(\frac{13}{20}\right)$ ampere, find the value of n. Assume that the cells have negligible internal resistance.



Ans. 3

Sol. $I = I_1 + I_2 + I_3 + I_4 = \frac{8}{10}-\frac{2}{8}+\frac{2}{4}-\frac{6}{2}$

$\frac{32-10+20-120}{40}=\frac{-78}{40}$ Amp



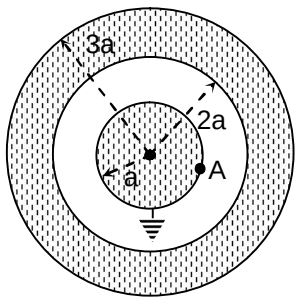
16. To disperse all the mass of the Earth to infinity against its own gravitational field would require 2.4×10^{32}J of energy. Mass of the earth is 6×10^{24} kg and radius of earth is 6400 km. Since earth is mostly iron, therefore, its specific heat capacity is about 500J kg^{-1} K^{-1}. Assuming that the earth was formed by very cold objects falling from a great distance away, if its maximum possible temperature soon after its formation is $\eta \times 10^4$ K. Find the value of η

Ans. 8

Sol. $2.4\times10^{32} = 6\times10^{24}\times500\times T$

$\Rightarrow \quad T=\frac{.4\times10^{8}}{500}=\frac{4\times10^{5}}{5}=8\times10^{4}K$

17. A conducting sphere of radius a is in side a hollow conducting sphere of radii 2a and 3a as shown in the figure. The inner sphere is earthed. The capacitance of system of sphere and hollow sphere is $5\pi k\varepsilon_0 a$. Find the value of k.

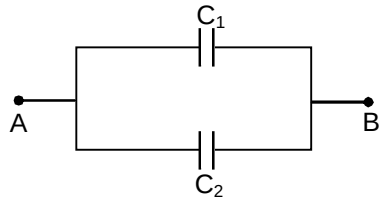


Ans. 4

Sol. $C_1 = 12\pi\varepsilon_0 a$

$$C_2 = \frac{4\pi\varepsilon_0 (b \times a)}{b - a} = 8\pi\varepsilon_0 a$$

$C_{eq} = C_1 + C_2 = 20\pi\varepsilon_0 a$



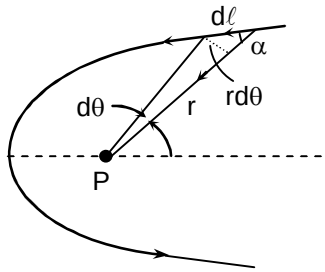
18. A current I = 20 ampere flows through a wire shaped in the form of an infinite parabola of latus rectum 4a (a = 1 mm). If the magnetic field at the focus of the parabola is $\eta\pi\times10^{-\beta}$ tesla, where η and β are the positive integer, less than 9. Find the value of $\eta\times\beta$.

Ans. 6

Sol. The magnetic field at the focus P of the parabola is given by the Biot-Savart law:

$$\vec{B} = \frac{\mu_0}{4\pi}\int \frac{I d\vec{\ell}\times\vec{r}}{r^3},$$

Where $\vec{r}$ and $d\vec{l}_0$ are as shown in figure. The direction of $\vec{B}$ is perpendicular to the plane of the parabola and out of it. We note that



$$\frac{\left|d\vec{\ell}\times\vec{r}\right|}{r^3} = \frac{r\,d\ell\sin\alpha}{r^3} = \frac{r\,d\theta}{r^2} = \frac{d\theta}{r}.$$

Therefore, $B = (\mu_0 I/4\pi)\int_0^{2\pi}(d\theta/r)$. The equation of the parabola is $r(1-\cos\theta) = 2a$, so that

$$B = \frac{\mu_0 I}{4\pi}\int_0^{2\pi}\frac{(1-\cos\theta)}{2a}d\theta = \frac{\mu_0 I}{4a} = \frac{4\times\pi\times10^{-7}\times20}{4\times10^{-3}} = 2\pi\times10^{-3} \text{ tesla}$$

Chemistry

PART – II

SECTION – A
(More Than One Correct Type)

This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

19. Which of the following statement(s) is/are incorrect?

(A) Major part of energy needed for the heterolysis of C – X bond in S_N1 reaction mechanism is obtained by the solvation of X^- by polar protic solvent.

(B) Greater solvation of attacking nucleophile and poor solvation of transition state promotes the reaction by S_N2 mechanism.

(C) Increase in the number of phenyl ring at C of C – X causes a shift in mechanism from S_N1 to S_N2.

(D) Allyl and benzyl halides shows both S_N1 and S_N2 reaction.

Ans. B, C

20. Select the correct statements:

(A) $NH_2 - NH_2$ does not give Lassaigne's test for detection of nitrogen.

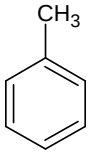
(B) HCOOH gives (+ve) $HgCl_2$ test.

(C) α-D-glucopyranosyl-β-D-fructofuranose shows muta rotation.

(D) $NH_2 - \overset{O}{\overset{\|}{C}} - NH - \overset{O}{\overset{\|}{C}} - NH_2$ gives Biuret test.

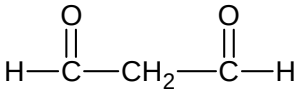
Ans. A, B, D

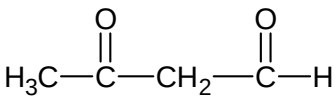
Sol. $HgCl_2 + HCOOH \xrightarrow{\Delta} \underset{(\text{Grey deposit})}{Hg} + CO_2 + 2HCl$

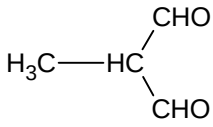
21. 

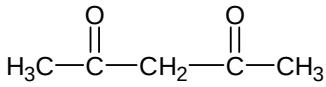
$\xrightarrow[\text{Liq.}NH_3]{Na} (A) \xrightarrow[Zn]{O_3} \text{Product}$

Which of the following product can be obtained as major product(s) in above reaction?

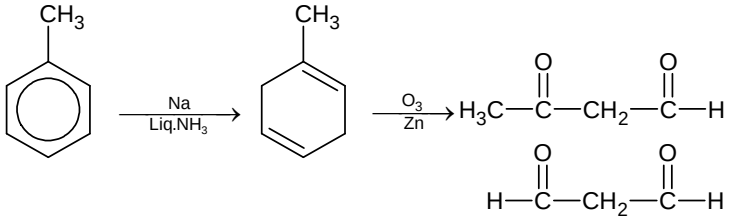
(A) 

(B) 

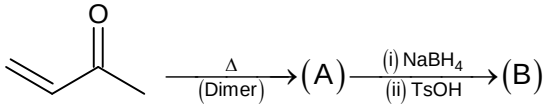
(C) 

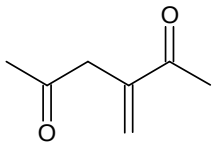
(D) 

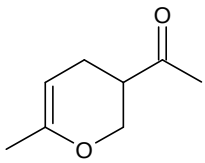
Ans. A, B

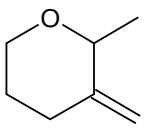
Sol. 

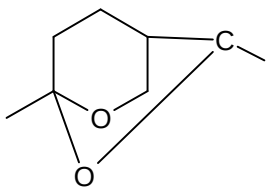
22. The major product A and B in the following reaction sequence are:



(A) 

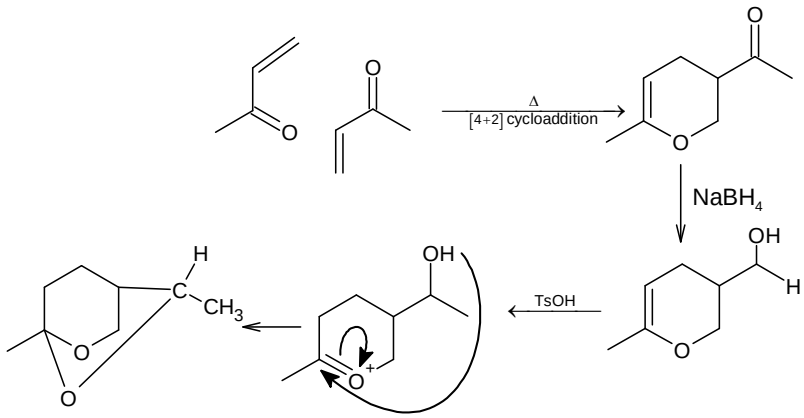
(B) 

(C) 

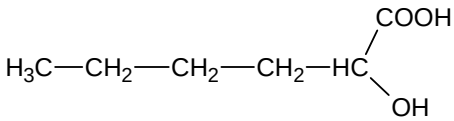
(D) 

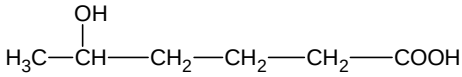
Ans. B, D

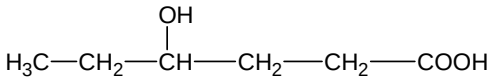
Sol.

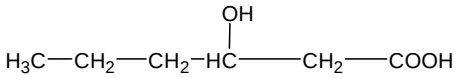


23. Which of the followings will give cyclic product upon heating?

(A) 

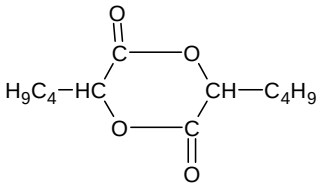
(B) 

(C) 

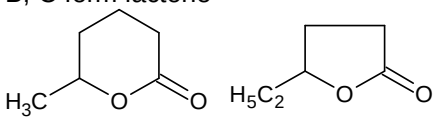
(D) 

Ans. A, B, C

Sol. A will form lactide



B, C form lactone



24. Which of the following are correct statements?

(A) In carboxylate, nucleophile prefer to attack on acyl carbon while in sulfonates nucleophile prefer to attack on alkyl carbon.

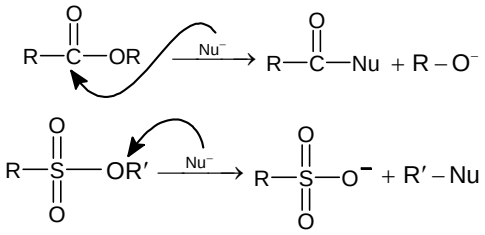
(B) Carboxylate anion is stronger base than sulfonate anion.

(C) Sulfonate anion is a better leaving group than carboxylate anion.

(D) Carboxylate anion is a better leaving group than sulfonate anion.

Ans. A, B, C

Sol.



Sulfonoic acid is stronger acid than carboxylic acid.

25. Which of the following statement(s) is/are correct?

(A) Sucrose has a pyranose and a furanose ring hold together by glycosidic linkage.

(B) Maltose has one glycosidic linkage and free hemiaccetal form

(C) Glucose, fructose and galactose forms same osazone

(D) ΔG^o for reaction, $\alpha - D - Glucose \rightleftharpoons \beta - D - Glucose$, is positive.

Ans. A, B

Sol. $\alpha - D - Glucose \rightleftharpoons \beta - D - Glucose$

36% 64%

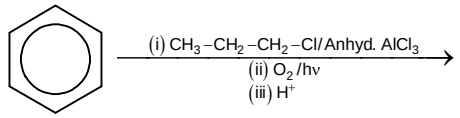
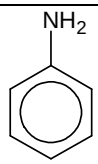
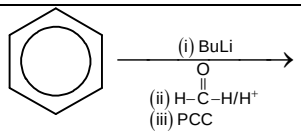
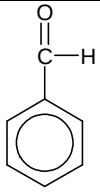
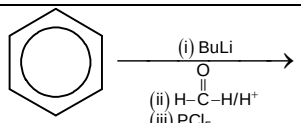
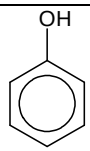
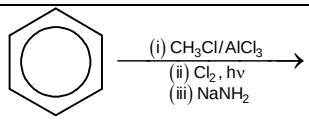
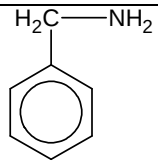
$$K_{eq} = \frac{64\%}{36\%} = \frac{16}{9} = 1.77$$

$$\Delta G^o = -2.303RT \log k_{eq} = -ve$$

(Matching Type)
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer Q. 26, Q. 27 and Q. 28 by appropriately matching the information given in the three columns of the following table.

Column 1		Column 2		Column 3	
(I)		(i)		(P)	Give ppt. when treated with Br_2/H_2O
(II)		(ii)		(Q)	In presence of strong base self disproportionation reaction takes place
(III)		(iii)		(R)	Form foul smelling compound with $CHCl_3$/NaOH
(IV)		(iv)		(S)	Form acetyl derivative with $R-\overset{O}{\overset{\|}{C}}-Cl$ / Pyridine

26. Which of the following combination is CORRECT for **I** ?

(A) (I) (ii) (R)

(B) (I) (iii) (P)

(C) (I) (ii) (P)

(D) (I) (iii) (Q)

Ans. B

27. Which of the following combination is CORRECT?

(A) (II) (ii) (Q)

(B) (II) (ii) (R)

(C) (III) (ii) (R)

(D) (IV) (i) (Q)

Ans. A

28. Which of the following combination is CORRECT for **IV** ?

(A) (IV) (i) (P)

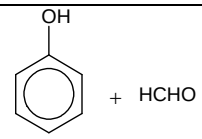
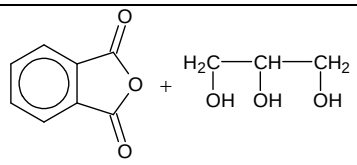
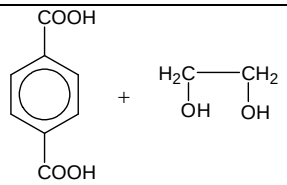
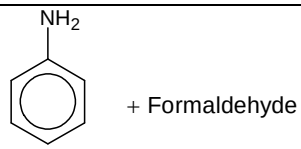
(B) (IV) (iv) (R)

(C) (IV) (i) (R)

(D) (IV) (i) (Q)

Ans. B

Answer Q. 29, Q. 30 and Q. 31 by appropriately matching the information given in the three columns of the following table.

Column 1		Column 2		Column 3	
(I)	Novolac	(i)		(P)	Cross linked
(II)	Bakelite	(ii)		(Q)	Linear chain
(III)	Dacron	(iii)		(R)	Branched chain
(IV)	Glyptal	(iv)		(S)	Fibre

29. Which of the following combination is CORRECT for Glyptal?

(A) (IV) (ii) (P)

(B) (IV) (ii) (Q)

(C) (IV) (iv) (P)

(D) (IV) (i) (P)

Ans. A

30. Which of the following combination is CORRECT for Bakelite?

(A) (II) (i) (Q)

(B) (II) (i) (P)

(C) (II) (iv) (R)

(D) (II) (iv) (P)

Ans. B

31. Which of the following combination is CORRECT?

(A) (IV) (iii) (P)

(B) (I) (iv) (P)

(C) (III) (iii) (S)

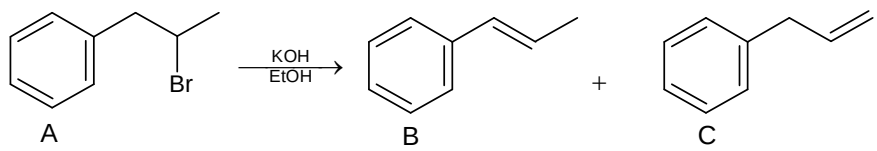
(D) (II) (iii) (S)

Ans. C

SECTION – C
Single Integer Answer Type

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

32.



A B C

1.99 gram of A on above reaction produces 1.062 gram mixture of B and C ratio of 2 : 1. if percentage of B formed will be x % then find the value of x/10.

Ans. 6

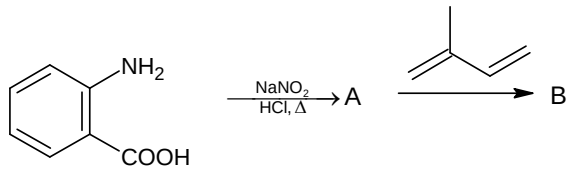
Sol. The theoretical yield will be $= \frac{118 \times 1.99}{199} = 1.18 \text{ gram}$

Hence percentage yield will be $= \frac{1.062}{1.18} \times 100\% = 90\%$

Hence, B formed will be $= 90\% \times \frac{2}{3} = 60\%$

x = 6.

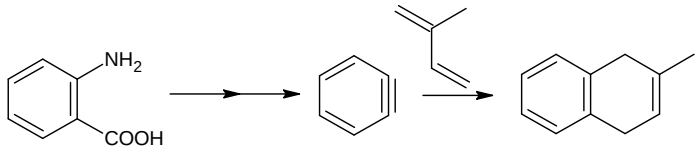
33.



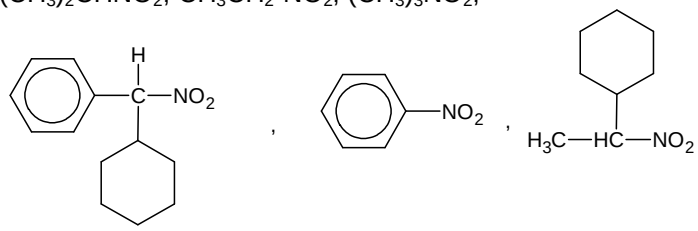
What is the double bond equivalent of B.

Ans. 6

Sol.

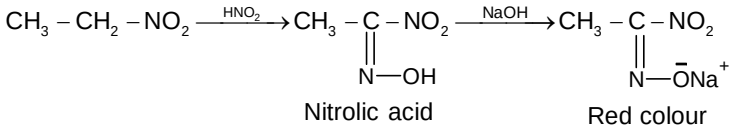


34. How many of given below compounds will form red colour with HNO_2 and followed by addition of NaOH?
$(CH_3)_2CHNO_2$, CH_3CH_2-NO_2, $(CH_3)_3NO_2$,



Ans. 1

Sol.

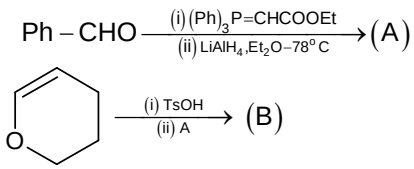


35. Find the number of nucleophile whose nucleophilicity is greater than pyridine in S_N2 reaction
$RS^-, ArS^-, I^-, CN^-, OH^-, F^-, NO_3^-$

Ans. 5

Sol. $RS^-, ArS^-, I^-, CN^-, OH^-$

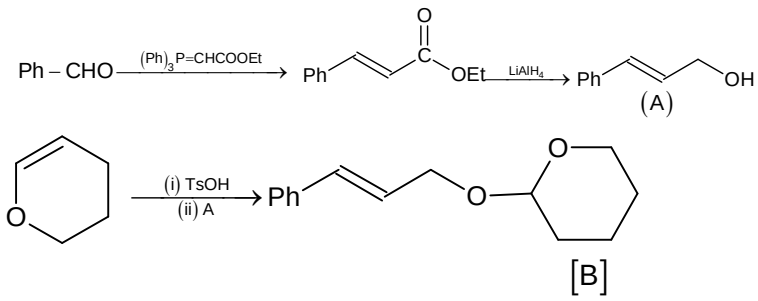
36.



How many oxygen atoms are there in B.

Ans. 2

Sol.



Mathematics

PART – III

SECTION – A

(More Than One Correct Type)

This section contains **7 multiple** choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct.

37. If obtuse angle bisectors of the lines $\tan^2 bx + 4y + 7 = 0$ and $x + \frac{\tan b}{4}y + 2 = 0$; $\left(b \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)\right)$ is $\frac{\tan^2 bx + 4y + 7}{\sqrt{\tan^4 b + 16}} = \frac{-\left(x + \frac{\tan b}{4}y + 2\right)}{\sqrt{1 + \frac{\tan^2 b}{16}}}$, then b can lie in the interval

(A) $\left(-\frac{\pi}{4}, -\frac{\pi}{12}\right)$

(B) $\left(-\frac{\pi}{4}, 0\right)$

(C) $\left(-\frac{\pi}{12}, -\frac{\pi}{18}\right)$

(D) $\left(\frac{\pi}{12}, \frac{\pi}{4}\right)$

Ans. **A, B, C**

Sol. $a_1a_2 + b_1b_2 < 0$

$\therefore \tan^2 b + \tan b < 0$

$-\frac{\pi}{4} < b < 0$

38. The global minimum value of

$|\cot x - 1| + |\cot x - 2| + |\cot x - 31| + |\cot x - 32| + |\cot x - 24| + |\cot x - 5| + |\cot x - 6|$
$+ |\cot x - 17| + |\cot x - 8| + |\cot x - 9| + |\cot x - 10| + |\cot x - 11| + |\cot x - 12|$

occurs at $x = \sec^{-1}\left(\frac{\alpha}{\beta}\right)$, then **INCORRECT** statements is/are (where [.] denotes the greatest integer function)

(A) $\sqrt{\alpha} + \beta = 31$

(B) $[\alpha + \beta] = 18$

(C) $\left[\frac{\alpha}{\beta}\right] = 2$

(D) $\left[\frac{\alpha^2}{\beta}\right] = 11$

Ans. **A, B, C, D**

Sol. Minimum sum occurs at cot x = 10

$\therefore x = \sec^{-1}\left(\frac{\sqrt{101}}{10}\right)$

39. If $f(x) = \sin^{-1}\left(\frac{4x}{4+x^2}\right)$, then

(A) $f\left(\sqrt{8}-2\right) = \pi - 2\tan^{-1}\left(\sqrt{2}-1\right)$

(B) $f\left(\sqrt{12}-\sqrt{3}\right) = \frac{2\pi}{3}$

(C) $f\left(-\frac{3}{\sqrt{8}-2}\right) = -\pi + 2\tan^{-1}\left(\frac{3}{\sqrt{8}-2}\right)$

(D) $f(\pi) = \pi - 2\tan^{-1}\left(\frac{\pi}{2}\right)$

Ans. **A, B, C, D**

Sol. $\sin^{-1}\left(\frac{4x}{4+x^2}\right) = \begin{cases} -\pi - 2\tan^{-1}\left(\frac{x}{2}\right) & ; \quad x < -2 \\ 2\tan^{-1}\left(\frac{x}{2}\right) & ; \quad -2 \le x \le 2 \\ \pi - 2\tan^{-1}\left(\frac{x}{2}\right) & ; \quad x > 2 \end{cases}$

40. Minimum value of $\sqrt{2x^2+2x+1} + \sqrt{2x^2-10x+13}$ is $\sqrt{\alpha}$, then

(A) α is an even number

(B) sum of digits of α is 2

(C) number of prime factors of α is 2

(D) number of prime factors is 3

Ans. **A, B, C**

Sol. $\sqrt{(\alpha+1)^2 + (\alpha+2-2)^2} + \sqrt{(\alpha-2)^2 + (\alpha+2-5)^2}$

Line is y = x + 2 and P(α, α + 2) is any point on this line and A ≡ (2, 5) and B(–1, 2)

41. Let ABC be a triangle and D be the mid-point of side BC. Suppose $\angle DAB = \angle BCA$ and $\angle DAC = 15^o$, then

(A) $\angle ADC$ is obtuse

(B) $\angle ADC$ is acute

(C) if O is the circumcentre of ΔADC, then ΔAOD is equilateral

(D) if O is the circumcentre of ΔADC, then triangle AOD is isosceles

Ans. A, C

Sol. $\frac{AD}{\sin B} = \frac{BD}{\sin\alpha}, \frac{CD}{\sin(15^o)} = \frac{AD}{\sin\alpha}$

Finally, we get $\angle ADC$ is obtuse

42. A hexagon inscribed in a circle has three consecutive sides of length 3 and three consecutive sides of length 5, and then the radius of the circle is

(A) $\frac{7}{\sqrt{3}}$

(B) $\frac{5}{\sqrt{2}}$

(C) $2\sqrt{3}$

(D) 4

Ans. A

Sol. Let the hexagon be ABCDEF with AB = BC = CD = 3 and DE = EF = FA = 5
So, $\angle AOB = \angle BOC = \angle COD = \alpha$ and $\angle DOE = \angle EOF = \angle FOA = \beta$
So, $\alpha + \beta = 120^o$

Now, $3 = 2r\sin\left(\frac{\alpha}{2}\right)$ and $5 = 2r\sin\left(\frac{\beta}{2}\right)$

$5 = 2r\sin\left(\frac{120^o - \alpha}{2}\right)$; $5 = r\left(\sqrt{3\left(1 - \frac{9}{4r^2}\right)} - \frac{3}{2r}\right)$; $r = \frac{7}{\sqrt{3}}$

43. If $S = \sum_{\beta=1}^{20}\sum_{\alpha=1}^{20}\tan^{-1}\left(\frac{\alpha}{\beta}\right)$, then S is equal to

(A) $100\,\pi$

(B) $300\,\pi$

(C) $400\,\pi$

(D) $225\,\pi$

Ans. A

Sol. $S = \sum_{\beta=1}^{20}\sum_{\alpha=1}^{20} \tan^{-1}\left(\frac{\alpha}{\beta}\right) = \left(20 \times \frac{\pi}{4}\right) + \frac{\pi}{2} \times 190 = 100\pi$

(Matching Type)
(Matching type - Single Correct Option)

This section contains **SIX** questions of matching type. The section contains **TWO** tables (each having 3 columns and 4 rows). Based on each table, there are **THREE** questions. Each question has **FOUR** options (A), (B), (C), and (D). **ONLY ONE** of these four options is correct.

Answer Q. 44, Q. 45 and Q. 46 by appropriately matching the information given in the three columns of the following table.

Column-1: contains the curve; Column-2: number of irrational points on the curve; Column-3: number of integral points on the curve (irrational points means both coordinates are irrational and integral points means both coordinates are integer)

Column-1	Column-2	Column-3
(I) $\frac{x^2}{9} + \frac{y^2}{4} = 1$	(i) infinite	(P) 2
(II) $\frac{x^2}{35} + y^2 = 1$	(ii) 800	(Q) 4
(III) $x^2 - y^2 = (200)^2$	(iii) 200	(R) 30
(IV) $x^4 - y^4 = 2401$	(iv) 16	(S) 100

44. Which of the following combination is **CORRECT**?

(A) (IV) (i) (P)

(B) (IV) (i) (R)

(C) (III) (i) (Q)

(D) (IV) (ii) (S)

Ans. A

45. Which of the following combination is **CORRECT**?

(A) (III) (i) (Q)

(B) (III) (ii) (R)

(C) (III) (i) (R)

(D) (I) (i) (P)

Ans. C

46. Which of the following combination is **INCORRECT**?

(A) (IV) (i) (P)

(B) (III) (i) (Q)

(C) (II) (i) (P)

(D) (IV) (ii) (S)

Ans. D

Sol. (II) For $\frac{x^2}{35}+y^2=1$, $\left(\sqrt{35}\cos\theta, \sin\theta\right)$

Let $\tan\theta = k\sqrt{35} \equiv \left(\frac{\sqrt{35}}{\sqrt{1+35k^2}}, \frac{k\sqrt{35}}{1+35k^2}\right)$

Let $\sqrt{\frac{1+35k^2}{35}} = k'$, $k'^2 - k^2 = \frac{1}{35}$

$(k'-k)(k'+k) = \frac{1}{35}$ infinite possible ways

(III) For $x^2 - y^2 = (200)^2$, $(x-y)(x+y) = 2^4 \times 5^4$

Let $(x > 0, y > 0)$ assign one 2 to each $(x+y)$ and $(x-y)$

Now, number of ways = $\frac{3\times 5-1}{2} = 7$

So, total integral points = 30

(IV) Number of integral points on $x^4 - y^4 = 2401$ is 2

Answer Q.47, Q.48 and Q.49 by appropriately matching the information given in the three columns of the following table.

Column-1: contains conics; Column-2: number of common tangents; Column-3: number of common normals

Column-1	Column-2	Column-3
(I) $2y^2 = 2x - 1$ and $2x^2 = 2y - 1$	(i) 0	(P) 0
(II) $(y-1)^2 = \frac{4}{\sqrt{3}}\left(x+\frac{\sqrt{3}}{4}\right)$ and $\frac{\left(x+\frac{\sqrt{3}}{4}\right)^2}{9} + \frac{(y-1)^2}{3} = 1$	(ii) 1	(Q) 1
(III) $x^2 = y - 2$ and $y + x^2 - 2x + 3 = 0$	(iii) 2	(R) 2
(IV) $y^2 = -4x$ and $x^2 = -4y$	(iv) 3	(S) 3

47. Which of the following options is the only **CORRECT** combination?

(A) (I) (iv) (Q)

(B) (I) (iv) (S)

(C) (II) (i) (S)

(D) (II) (ii) (S)

Ans. A

48. Which of the following options is the only **CORRECT** combination?

(A) (IV) (i) (Q)

(B) (IV) (i) (P)

(C) (IV) (ii) (Q)

(D) (I) (iv) (P)

Ans. C

49. Which of the following options is the only **INCORRECT** combination?

(A) (III) (iii) (Q)

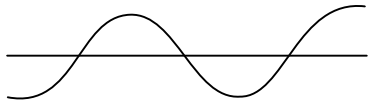
(B) (II) (i) (S)

(C) (II) (iii) (S)

(D) (IV) (ii) (Q)

Ans. B

Sol. (I) $2y^2 = 2x - 1$ and $2x^2 = 2y - 1$



Equation of tangents are $y = m\left(x - \frac{1}{2}\right) + \frac{1}{4m}$

and $y - \frac{1}{2} = mx - \frac{m^2}{4}$ respectively

So, $-\frac{m}{2} + \frac{1}{4m} = \frac{1}{2} - \frac{m^2}{4}$; $m^3 - 2m^2 - 2m + 1 = 0$

This equation has 3 real and distinct roots i.e., 3 common tangents

For common normals they are symmetric about y = x

So, only one common normal

(II) After shifting at origin, we get $y^2 = \frac{4}{\sqrt{3}}x$ and $\frac{x^2}{9} + \frac{y^2}{3} = 1$

Equation of normals are $y = mx - \frac{2}{\sqrt{3}}m - \frac{1}{\sqrt{3}}m^3$ and $y = mx \pm \frac{m\left(a^2 - b^2\right)}{\sqrt{a^2 + b^2m^2}}$

So, $\frac{2}{\sqrt{3}}m + \frac{1}{\sqrt{3}}m^3 = \frac{6m}{\sqrt{9 + 3m^2}}$; m = 0, ±1. Hence, 3 common normals

(III) $y = x^2$ and $y = -(x - 1)^2$ equation of tangents are $y = mx - \frac{m^2}{4}$

and $y = m(x - 1) + \frac{m^2}{4}$ respectively

So, we get m = 0, 2; 2 common tangents

Common normals: equation are $y = mx + \frac{1}{2} + \frac{1}{4m^2}$ and $y = mx - m - \frac{1}{4m^2} - \frac{1}{2}$

So, we get $2m^3 + 2m^2 + 1 = 0$ only one real root is 1 common normals

(IV) $y^2 = -4x$ and $x^2 = -4y$

Number of common tangents will be same

For $y^2 = 4x$ and $x^2 = 4y$ only one common tangents

Common normals: $2m + m^3 = -2 - \frac{1}{m^2}$

$f(m) = m^5 + 2m^3 + 2m^2 + 1$ only one root negative

So, only one common normals

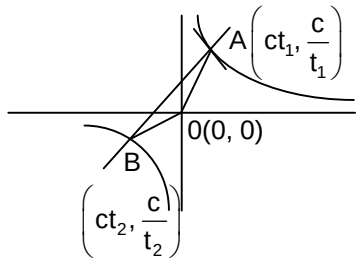
SECTION – C
(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

50. If a chord AB of the curve $y = \frac{3x-2}{x-2}$ is a normal to the curve at a point A, subtending an angle θ at the point P(2, 3). Then the value of $\frac{\sin(\theta - \angle PAB)}{\sin(\theta + \angle PAB)}$ is equal to _____

Ans. 3

Sol. Shift the curve so that centre is origin, $xy = 4$



$\because\ t_2 = \frac{-1}{t_1^3};\ m(OA) = \frac{1}{t_1^2}, m(OB) = \frac{1}{t_2^2} = t_1^6;\ \tan\theta - \frac{1-t_1^4}{t_1^2}$

Also, $\tan(\angle OAB) = \frac{t_1^2 - \frac{1}{t_1^2}}{1+1} = \frac{t_1^4 - 1}{2t_1^2}$

$\therefore\ \frac{\tan\theta}{\tan(\angle OAB)} = -2$

So, $\frac{\sin(\theta - A)}{\sin(\theta + A)} = 3$

51. Assuming that no three circles passing through same points and every two circles intersects in two distinct points. If the number of regions that points are created by 7 mutually non over-lapping circles in a plane is 11k, then k is _____

Ans. 4

Sol. $T_n = T_{n-1} + 2(n-1): n \geq 2$

$T_n = n^2 - n + 2$

$T_7 = 7^2 - 7 + 2 = 44$

52. The combined equation of the principal axes of the hyperbola $2x^2 + 12xy - 7y^2 - 16x + 2y - 5 = 0$ is $ax^2 + 2hxy + by^2 - x + 7y + c = 0$, then $|a + b - 2h + c|$ is equal to _____

Ans. 6

Sol. $c \equiv (1, 1)$; The axes are the bisectors of the angle between the asymptotes

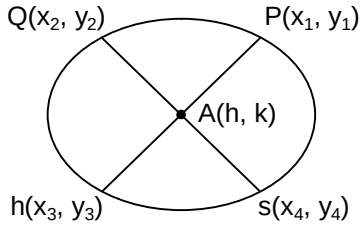
So, $\frac{(x-1)^2-(y-1)^2}{(x-1)(y-1)}=\frac{2-(-7)}{6}$

$2x^2 - 3xy - 2y^2 - x + 7y - 3 = 0$

53. From the point (1, 1) normals are drawn to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$, then locus of conormal points is $\lambda_1xy + \lambda_2x + \lambda_3y + \lambda_4x^2 + \lambda_5y^2 + \lambda_6 = 0$, then $\lambda_1 + \lambda_2 + \lambda_3 - \lambda_4 - \lambda_5 - \lambda_6$ is equal to _____

Ans. 0

Sol. Equation of normal at $P(x_1, y_1)$; $\frac{x-x_1}{\frac{x_1}{a^2}}=\frac{y-y_1}{\frac{y_1}{b^2}}$



$\Rightarrow (a^2 - b^2)x_1y_1 + b^2yx_1 - a^2xy_1 = 0$

Point (h, k) lies on it

So; $(a^2 - b^2)x_1y_1 + b^2kx_1 - a^2hy_1 = 0$

Similarly, for points Q, R and S are

$(a^2 - b^2)x_2y_2 + b^2kx_2 - a^2hy_2 = 0$

$(a^2 - b^2)x_3y_3 + b^2kx_3 - a^2hy_3 = 0$ and $(a^2 - b^2)x_4y_4 + b^2kx_4 - a^2hy_4 = 0$

Thus, P, Q, R, S lie on the curve $(a^2 - b^2)xy + b^2kx - a^2hy = 0$ i.e., $5xy + 4x - 9y = 0$

54. Three circles C_1, C_2 and C_3 with radii 2, r, 8 respectively are given. They are placed such that C_2 lies to the right of C_1 and touches it externally, C_3 lies to the right of C_2 and touches it externally. Further, there exist two straight lines each of which is a direct common tangent simultaneously to all three circles, then r is _____

Ans. 4

Sol. $r = \sqrt{2 \times 8}$